Test Data Generator 2.0

13 Oct 2026 · Stage paths like a real ART, extracts that move on, and structure learned from real data.

What is it?

The generator walked neatly through every stage: eleven transition pairs, no loops, a median of seven transitions per issue. A real ART (458 issues) shows forty pairs, twelve percent backward, seven percent self-loops, cancellations straight into Done, median four. And every run was a one-off – two extracts with time in between did not exist.

What it does now (0.38.0 to 0.41.0)

• Realism knob 0–100: eight disturbances (skips, cancels, parking, self-loops, long backflow, reopening, start outside the first stage, archiving); about 70 matches the real ART, calibration pinned by tests
• Continuation: --continue produces the second extract – the past byte for byte, open issues move on, new ones join, reproducible by seed and cut date
• Whole data room: the demo portfolio continues in place, configs move, snapshots rotate – the delta briefing shows real elapsed time
• Learning from real data: --learn-from estimates the transition matrix and dwell times from a transitions file – counts and hours only, no keys

Three decisions

Path before time. The stage sequence is drawn first, the time spread over it afterwards. That is why realism and profile act in every mode, flow patterns included, and why an open issue can simply get a new path at the cut.
The past is untouchable. A second extract changes nothing of what the first one showed – like two manual pulls from Jira. Tests compare the data byte for byte.
Structure yes, content no. The learned profile carries stage names, counts and hours. What comes out of it shares only the shape with its source.

How to use it

```
python -m testdata_generator --workflow wf.txt --issues 500 --realism 70
python -m testdata_generator --workflow wf.txt --continue ART.json --to-date 2026-03-01
python -m testdata_generator --scenario portfolio --continue demo/ --to-date 2026-03-01
python -m testdata_generator --learn-from ART_A_Transitions.xlsx --workflow wf.txt
```